

AI-Integrated Smart Intercity Bus Scheduling & E-Ticketing

Problem

Intercity bus ticketing in Ethiopia is still mostly manual. Passengers must go physically to terminals to check schedules and buy tickets. There is no centralized platform to compare bus companies, prices, routes, or seat availability. Travelers also face problems like sold-out tickets, ticket scalping, lack of seat selection, and paper ticket loss.

Proposed Solution

A web-based platform that allows users to:

- Search routes and schedules
- Compare bus companies and prices
- Select seats online
- Book digital QR-code tickets
- Manage bookings and trip history

The system will also include AI features such as:

- Smart bus recommendations
- Peak travel prediction
- Seat occupancy prediction
- Personalized travel suggestions

Technologies

- Frontend: React.js
- Backend: Node.js / Express.js
- Database: PostgreSQL
- AI Module: Python + Scikit-learn
- Authentication: JWT

Group Members

-Aman Baye

-Amanuel Neby

-Kindie Abera